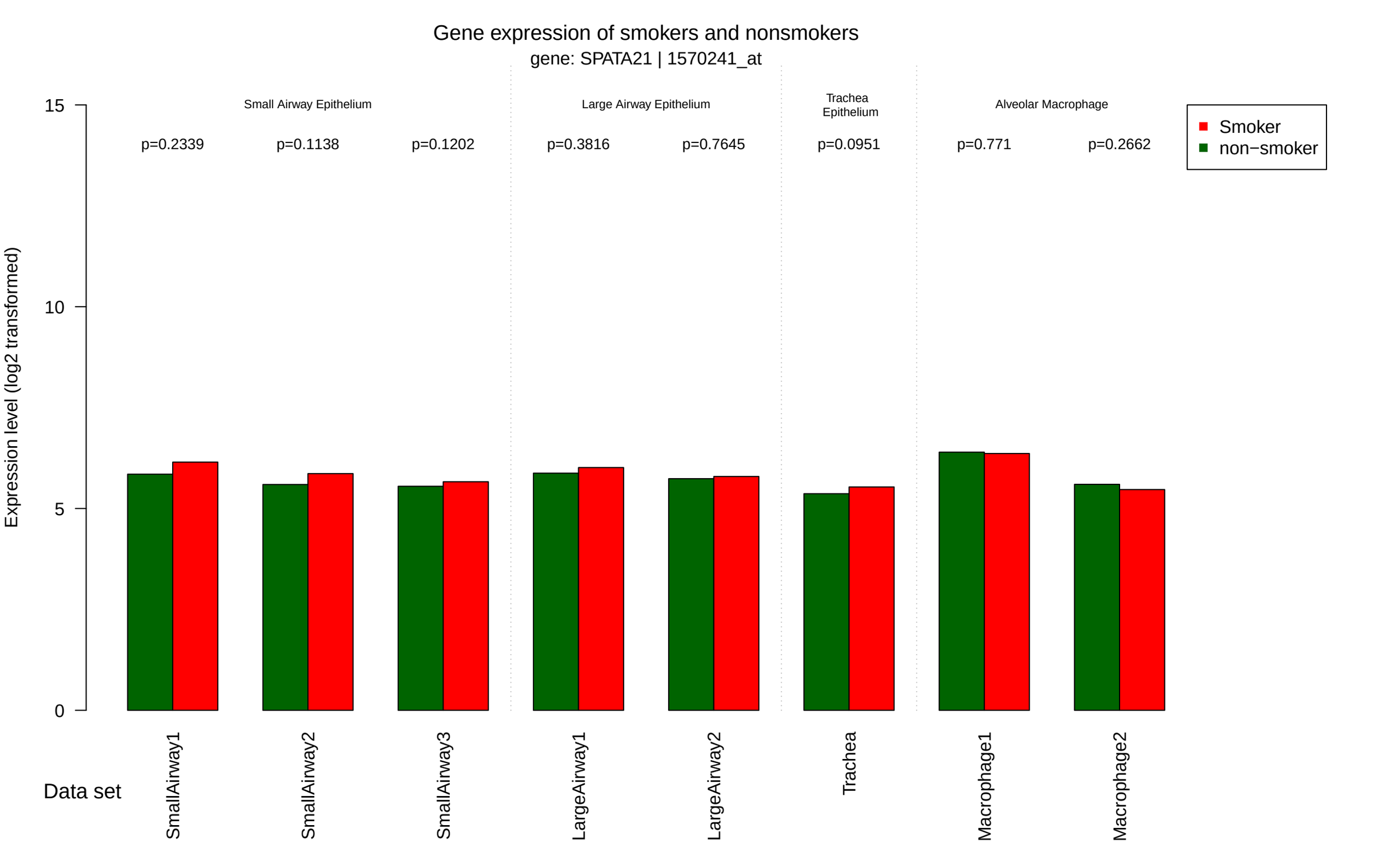
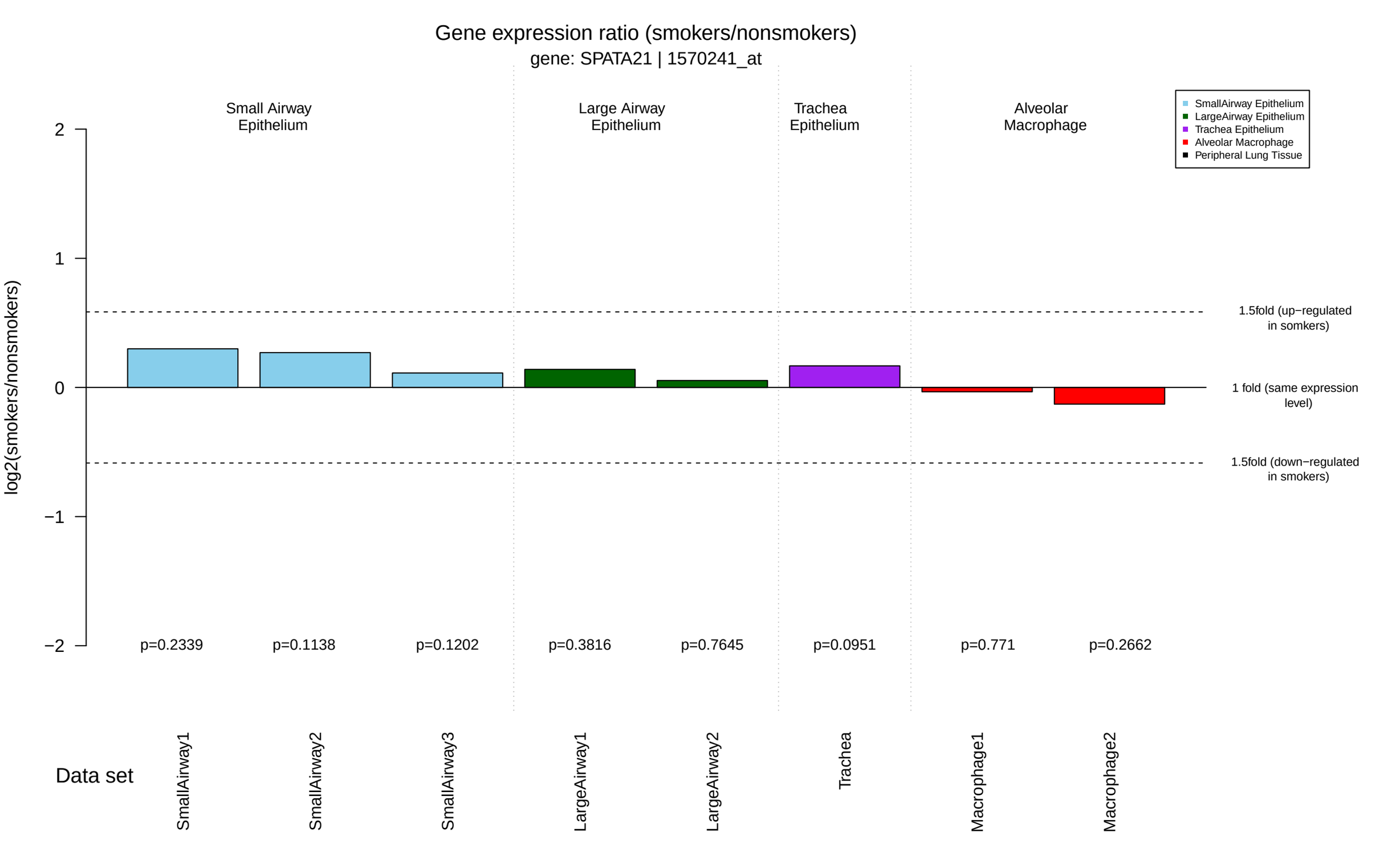
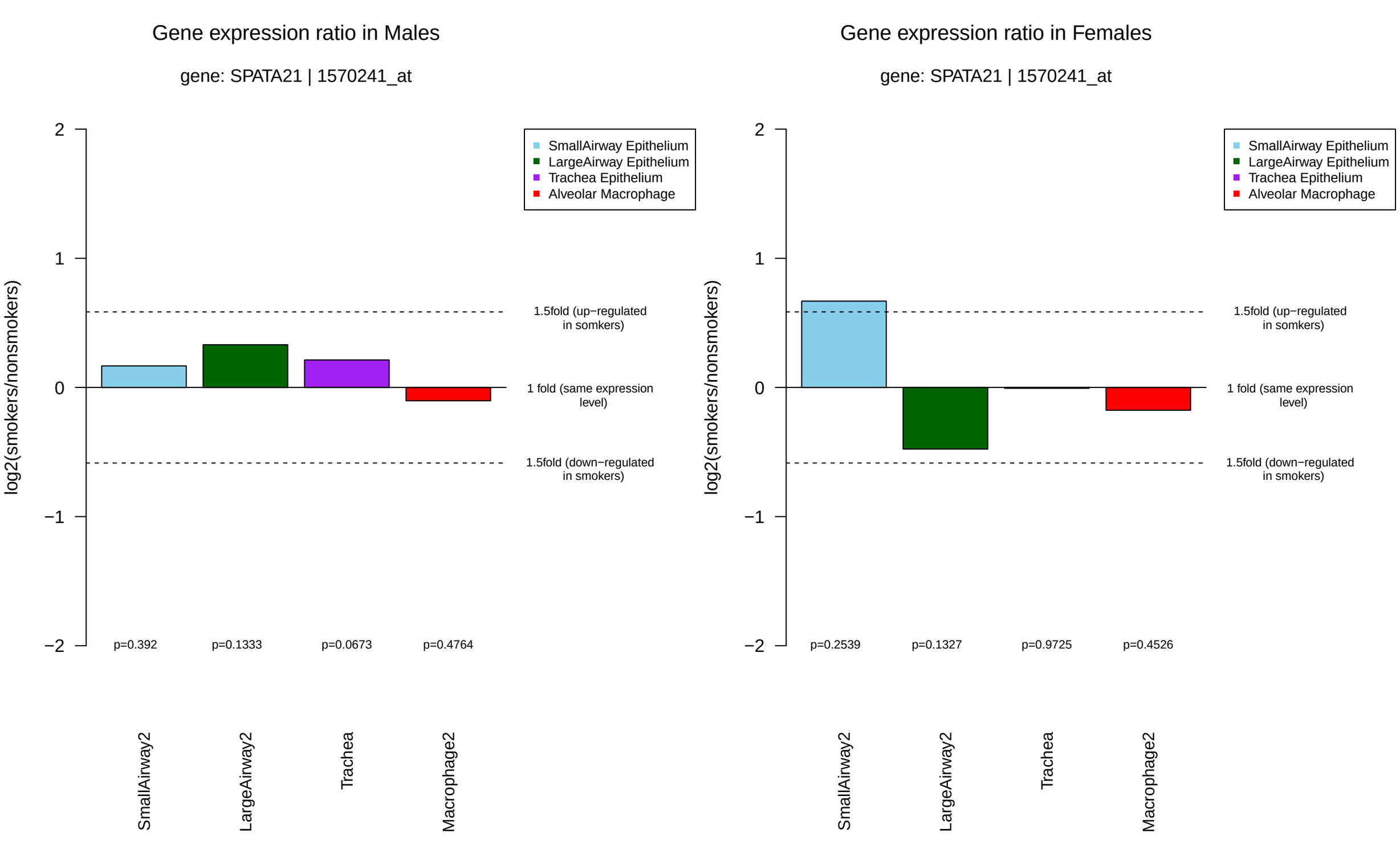
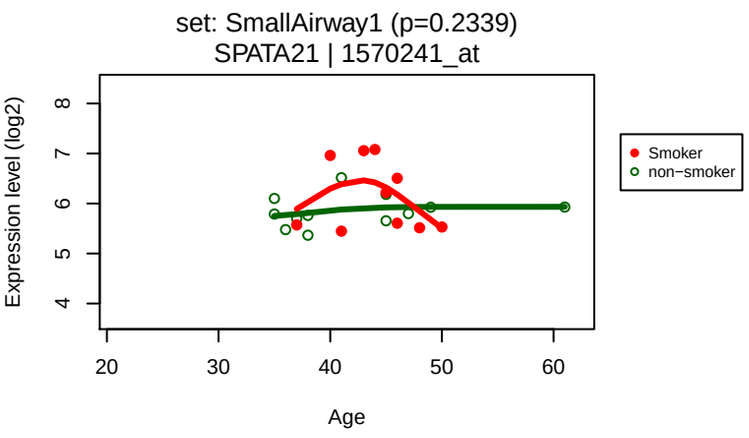
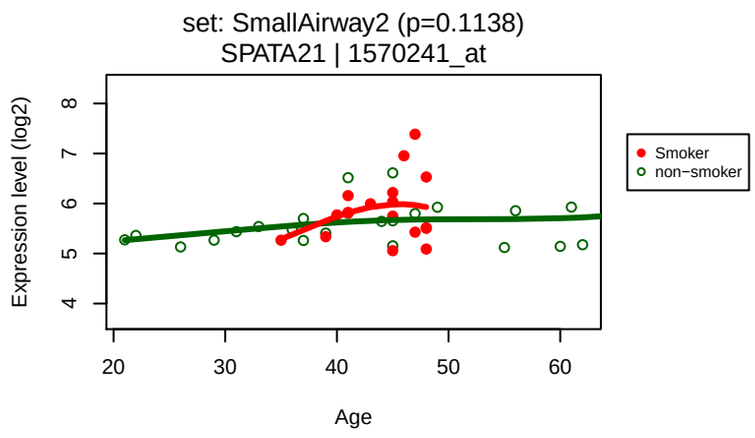
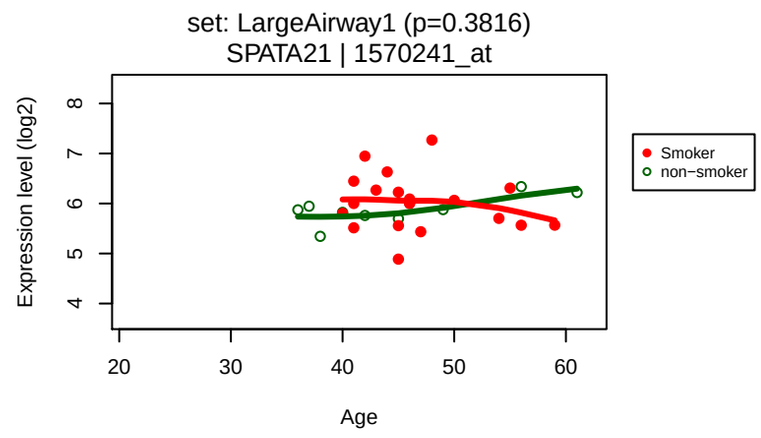

Gene expression of smokers and nonsmokers
gene: SPATA21 | 1570241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2339
p=0.1138
p=0.1202
p=0.3816
p=0.7645
p=0.0951
p=0.771
p=0.2662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

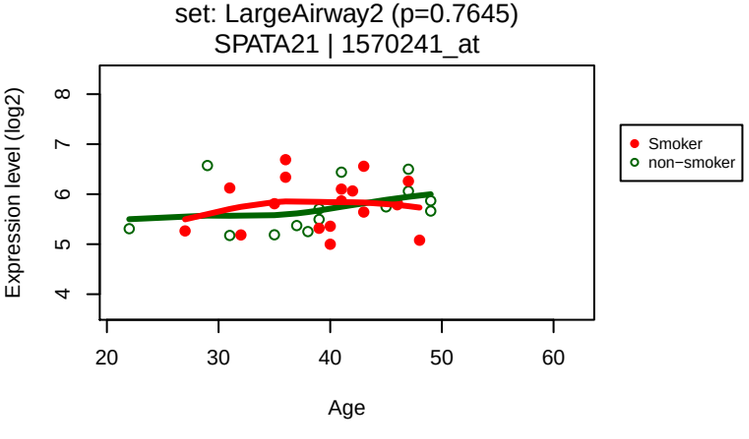

Gene expression ratio (smokers/nonsmokers)
gene: SPATA21 | 1570241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2339
p=0.1138
p=0.1202
p=0.3816
p=0.7645
p=0.0951
p=0.771
p=0.2662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

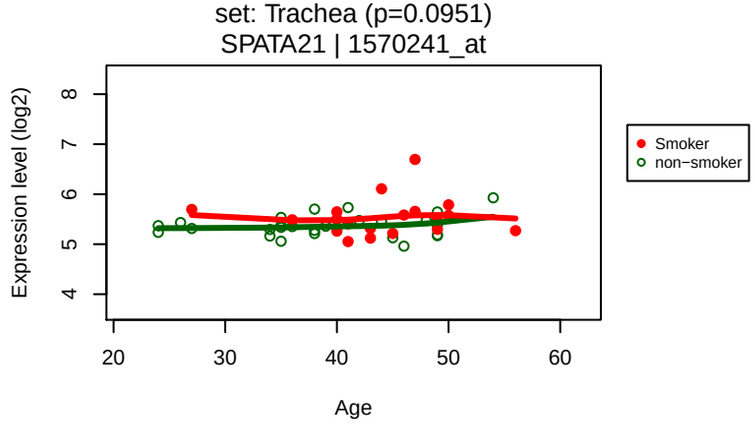

Gene expression ratio in Males
gene: SPATA21 | 1570241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.392
p=0.1333
p=0.0673
p=0.4764
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA21 | 1570241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.1327
p=0.9725
p=0.4526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

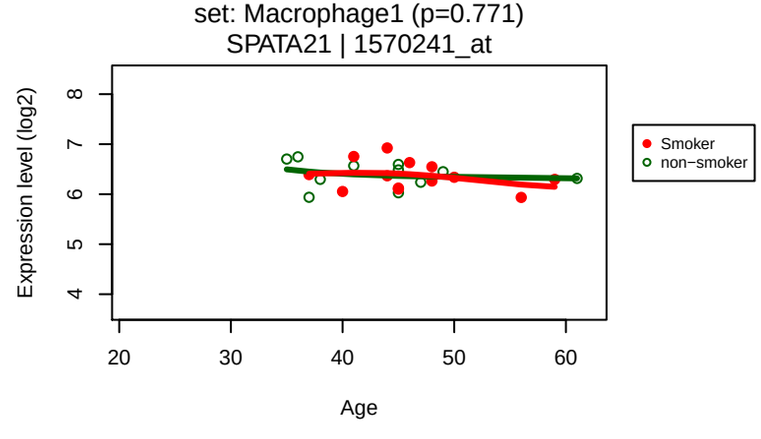

set: SmallAirway1 (p=0.2339)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker

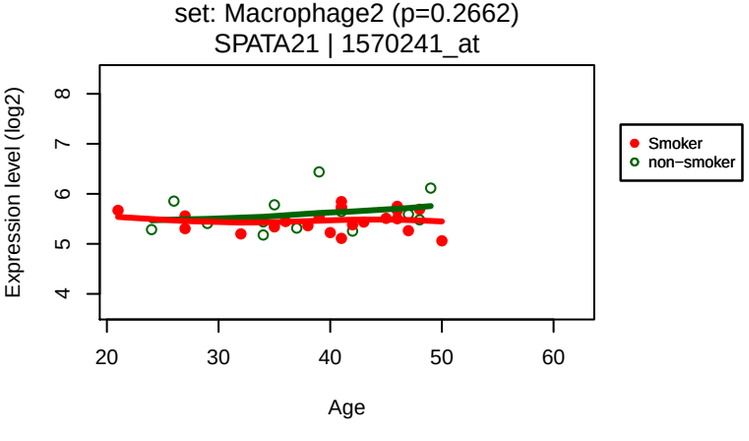

set: SmallAirway2 (p=0.1138)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker

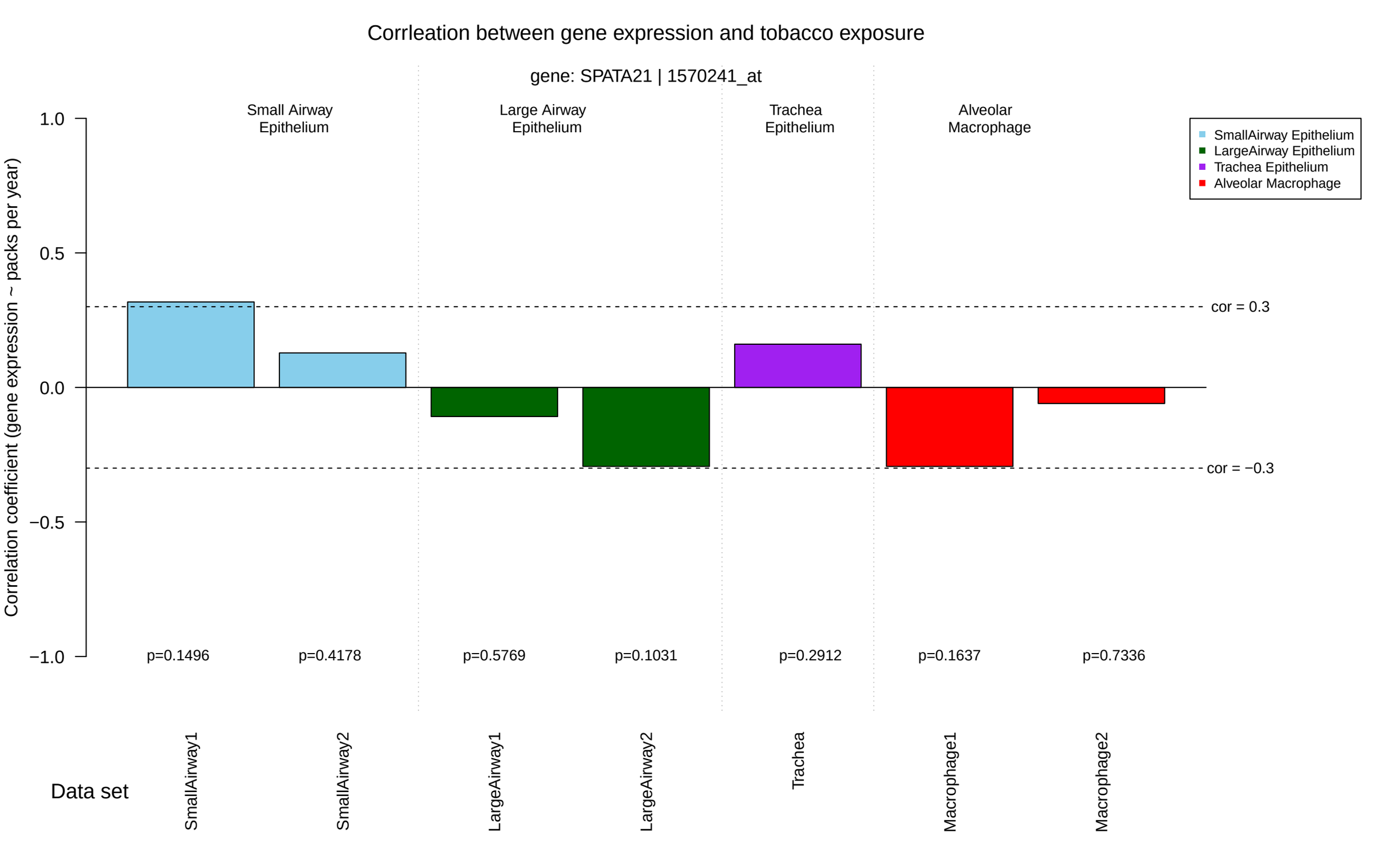

set: LargeAirway1 (p=0.3816)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7645)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0951)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.771)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2662)
SPATA21 | 1570241_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA21 | 1570241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1496
p=0.4178
p=0.5769
p=0.1031
p=0.2912
p=0.1637
p=0.7336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2